

Literature Review

The increasing focus on explainability in artificial intelligence (AI) has motivated research into multimodal systems capable of generating interpretable and pedagogically effective explanations. TheoremExplainAgent by Ku et al. (2025) presents a pioneering framework for producing long-form, video-based theorem explanations that integrate visual and textual reasoning. This work extends the frontier of AI interpretability by combining large language models (LLMs), agentic planning, and automated visualization through Manim-based animations. The system seeks to bridge the gap between symbolic reasoning and human comprehension by offering structured, multimodal explanations across multiple STEM disciplines.

The architecture of TheoremExplainAgent consists of two cooperative LLM agents: a planner agent that constructs a narrative and storyboard, and a coding agent that translates the plan into executable Manim scripts. This modular design reflects contemporary trends in agentic AI systems (Abramovich et al., 2024; He et al., 2024a), where models autonomously plan and execute tasks by interacting with tools and environments. The study also integrates a retrieval-augmented generation (RAG) mechanism, allowing the agents to dynamically access documentation, visual examples, and code templates to improve animation fidelity. Through these combined mechanisms, TheoremExplainAgent demonstrates that structured video generation can serve as an interpretable medium for communicating complex theorems.

To systematically evaluate multimodal reasoning, the authors introduce TheoremExplainBench, a benchmark comprising 240 theorems from mathematics, physics, chemistry, and computer science, spanning multiple difficulty levels. The benchmark is assessed through five quantitative metrics—accuracy and depth, visual relevance, logical flow, element layout, and visual consistency—enabling holistic evaluation of both factual and perceptual quality. Similar to prior multimodal reasoning benchmarks such as ScienceQA (Lu et al., 2022) and TheoremQA (Chen et al., 2023b), this dataset advances the methodological rigor for assessing AI comprehension, yet uniquely emphasizes multimodal synthesis rather than text-only reasoning.

Experimental results demonstrate that TheoremExplainAgent achieves significant improvements in long-form video generation compared to text-to-video systems like LTXVideo (HaCohen et al., 2024) and Veo2 (Veo-Team et al., 2024), which fail to maintain pedagogical structure or semantic alignment. The o3-mini agent attained a 93.8% success rate in generating complete, coherent videos—substantially outperforming models such as GPT-4o (OpenAI, 2023) and Gemini 2.0 Flash (DeepMind, 2025). Nonetheless, challenges persist in visual layout accuracy, consistency, and retrieval reliability, echoing broader issues identified in RAG-based code generation (Soman & Roychowdhury, 2024).

Beyond performance metrics, the study highlights the interpretive value of visual explanations. Through comparative analysis, Ku et al. (2025) show that video-based reasoning surfaces conceptual flaws—such as misapplied theorems or direction-encoding errors—that remain obscured in text. This aligns with cognitive theories of multimodal learning, which suggest that visual information enhances conceptual understanding by concretizing abstract ideas (Mayer, 2014). The findings thus reinforce the pedagogical importance of multimodal AI explainability.

In conclusion, TheoremExplainAgent represents a significant advancement in automated scientific communication, establishing a scalable framework for AI-generated, multimodal theorem understanding. By combining LLM-driven reasoning, agentic planning, and visual storytelling, the system not only enhances interpretability but also exposes hidden reasoning limitations in current models. This research lays foundational groundwork for future exploration into AI-driven education, video-based explainability, and multimodal reasoning evaluation, marking a step toward more transparent and human-aligned AI systems.

References

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